1. Method

Overview

In this work, I implemented a **weakly supervised semantic segmentation pipeline** for remote sensing images using **partial cross-entropy loss** with sparse point-level annotations.

Instead of using dense pixel-wise labels, the model is trained using **randomly sampled points per image**, simulating a realistic low-annotation scenario.

Model Architecture

- Backbone: ResNet (18, 34, 50 evaluated)
- Segmentation Head: Fully Convolutional Network (FCN-style)
- Input: RGB remote sensing images
- Output: Pixel-wise class predictions

Partial Cross-Entropy Loss

To handle sparse supervision:

- Only labeled pixels (sampled points) are used
- Unlabeled pixels are ignored using: ignore_index = -1
- The loss is computed only over valid labeled pixels, allowing the model to learn from incomplete annotations.

Data Preparation

- Dataset size: **41 images**
- Split into:
    - Training set
    - Validation set
    - Test set

- For each image:
  - A fixed number of pixels are randomly sampled
  - Sampled pixels retain their labels
  - Remaining pixels are set to **-1 (ignored)**

Limitation: Random Point Sampling

One important aspect of this work is that the labeled points are sampled **randomly**.

This introduces variability in the supervision signal:

- Different runs may observe different pixels
- Important regions or small classes may not be sampled
- Class imbalance may occur due to randomness

This explains why increasing the number of sampled points does not always improve performance.

2. Experiments

To evaluate the system, I designed experiments to study the effect of:

1. Number of sampled points
2. Backbone architecture
3. Number of training epochs

Experiment 1: Effect of Number of Points

Purpose

To evaluate how the number of labeled points per image affects segmentation performance.

Hypothesis

Increasing the number of points improves performance up to a limit, after which performance may degrade due to redundancy or noise.

Setup

- Model: ResNet34
- Epochs: fixed
- Points tested: 50, 100, 250, 500, 1000

Results

Points	Pixel Accuracy	mIoU
50	0.711	0.234
100	0.684	0.243
250	**0.806**	**0.417**
500	0.701	0.340
1000	0.702	0.246

Analysis

- Best performance achieved at **250 points**
- Increasing points beyond that **reduced performance**
- Likely reasons:
    - Redundant supervision
    - Noise from random sampling
    - Poor class distribution

Insight:

More supervision is not always better in weakly supervised learning.

Discussion

The results show that segmentation performance does not strictly increase with the number of sampled points.

This behavior can be explained by the **random sampling process**. Different random point configurations may provide varying levels of informative supervision. In some cases, a smaller number of well-distributed points may provide sufficient information for the model to learn segmentation boundaries.

However, overall performance remains stable across different sampling sizes, indicating that the model can learn meaningful segmentation patterns even with sparse supervision.

Experiment 2: Effect of Backbone Architecture

Purpose

To study the impact of model complexity on learning from sparse annotations.

Hypothesis

- Small models may underfit
- Large models may overfit
- Medium models may perform best

Setup

- Points: 250 (best from previous experiment)
- Models tested:
    - ResNet18
    - ResNet34
    - ResNet50
- Epochs tested: 10

Results

Model	Pixel accuracy	mIoU
Resnet18	0.704	0.20
Resnet34	0.806	**0.417**
Resnet50	**0.828**	0.350

Analysis

- ResNet34 achieved the best segmentation performance
- ResNet50:
    - Higher pixel accuracy
    - Lower mIoU → weaker boundary quality
- ResNet18 underfits the data

Insight:

Medium-depth models provide the best balance between capacity and generalization.

Experiment 3: Effect of Training Epochs

Purpose

To analyze how training duration affects performance.

Hypothesis

- Increasing epochs improves performance initially
- Too many epochs cause overfitting

Setup

- Model: ResNet50
- Epochs tested: 5, 10, 20
- Points: 250

Results

Epochs	Pixel accuracy	mIoU
5	0.733	0.249
10	**0.828**	**0.350**
20	0.729	0.261

- Best performance at **10 epochs**
- Performance drops at 20 epochs → overfitting

Insight:

Early stopping is important when training with sparse labels.

The results show that increasing the backbone capacity does not necessarily lead to better segmentation performance.

Discussion

The results indicate that training for **10 epochs** provides the best balance between learning and generalization.

Training for 20 epochs results in a significant drop in performance, which suggests **overfitting**. Since the model is trained using sparse point annotations, it may begin to memorize the limited labeled pixels instead of learning general segmentation patterns.

To better evaluate the model's performance, qualitative results are visualized for several test samples. Each row represents a different input image and contains four components:

- **Original Image**: The raw satellite image input to the model.
- **Ground Truth Mask**: The true segmentation map representing land cover classes.
- **Prediction**: The output generated by the model.
- **Overlay**: A combination of the predicted segmentation and the original image for better visual interpretation.

3. Visualization

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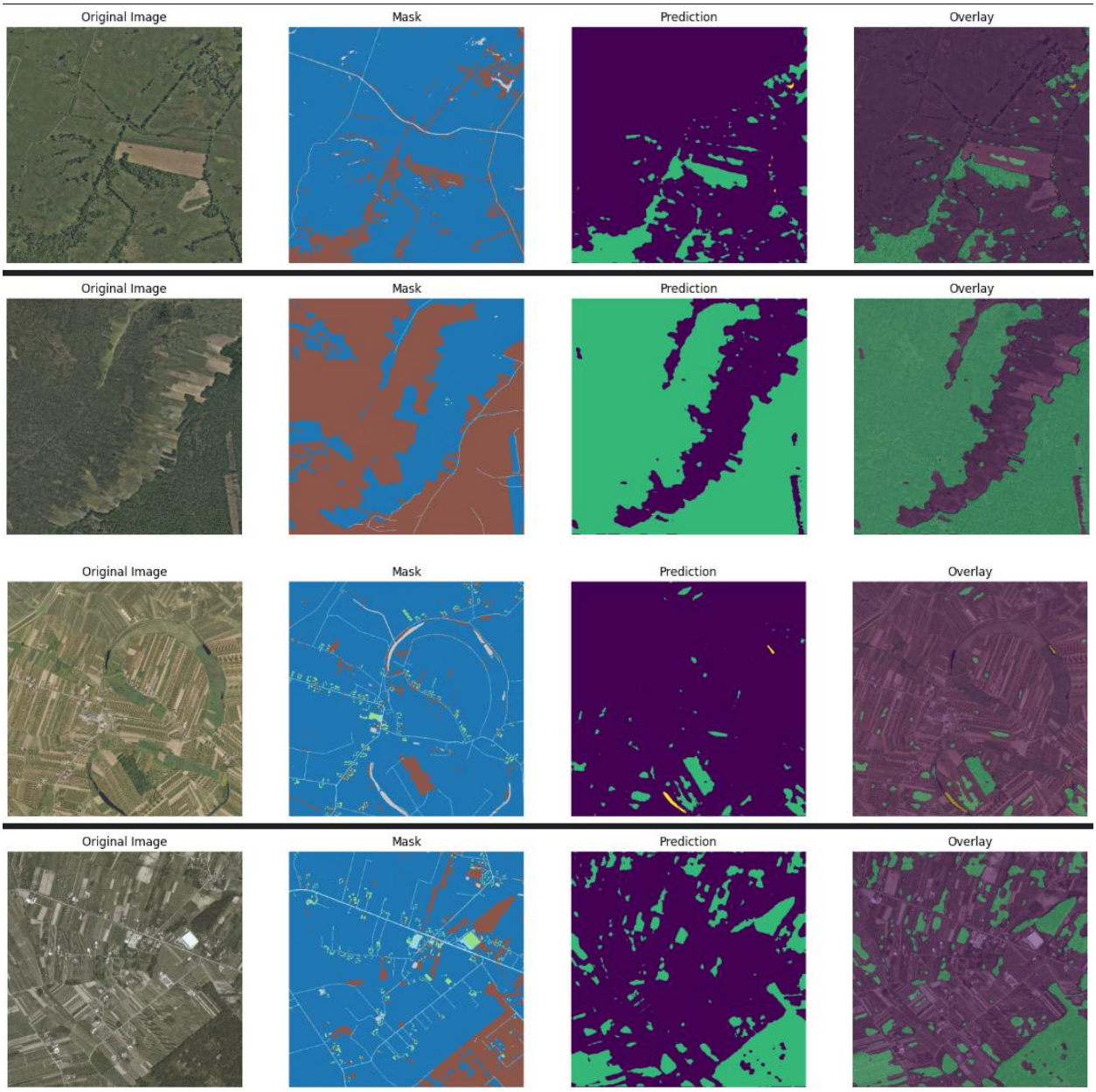

4. Conclusion

This work explored the use of **Partial Cross Entropy Loss** for training a remote sensing segmentation network with simulated point supervision.

Key findings:

- The model can learn meaningful segmentation patterns using only sparse point annotations.
- Increasing the number of labeled points does not always guarantee improved performance due to randomness in point sampling.
- Larger backbone architectures do not necessarily improve results under sparse supervision.
- Training duration must be carefully controlled to avoid overfitting.

Overall, the experiments demonstrate that **weakly supervised segmentation using point annotations can achieve reasonable performance while significantly reducing annotation cost.**